couldn't figure out how to eliminate it. Finally they looked inside the dish and saw pigeons nesting there. And so they were worried that a white dielectric substance (pigeon poop) might be responsible for the signal, because they detected it no matter what direction the detector pointed. After cleaning out the dielectric substance, the interference dropped a little bit, but a leftover signal remained. The paper they published in 1965 was all about this unaccountable "excess antenna temperature."††

Meanwhile, a team of physicists at Princeton, led by Robert Dicke, was building a detector specifically to find the CMB. But they didn't have the resources of Bell Labs, so their work went a little slower. And the moment Dicke and his colleagues heard about Penzias and Wilson's work, the Princeton team knew exactly what the observed excess antenna temperature was. Everything fit: especially the temperature itself, and that the signal came from every direction in the sky.

In 1978, Penzias and Wilson won the Nobel Prize for their discovery. And in 2006, American astrophysicists John C. Mather and George F. Smoot would share the Nobel Prize for observing the CMB over a broad range of the spectrum, bringing cosmology from a nursery of clever but untested ideas into the realm of a precision, experimental science.

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Because light takes time to reach us from distant places in the universe, if we look out in deep space we actually see eons back in time. So if the intelligent inhabitants of a galaxy far, far away were to measure the temperature of the cosmic background radiation at the moment captured by our gaze, they should get a reading higher than 2.7 degrees, because they are living in a younger, smaller, hotter universe than we are.

Turns out you can actually test this hypothesis. The molecule cyanogen CN (once used on convicted murderers as the active component of the gas administered by their executioners) gets excited by exposure to microwaves. If the microwaves are warmer than the ones in our CMB, they excite the molecule a little more. In the big bang model, the cyanogen in distant, younger galaxies gets bathed in a warmer cosmic background than the cyanogen in our own Milky Way galaxy. And that's exactly what we observe.

You can't make this stuff up.

Why should any of this be interesting? The universe was opaque until 380,000 years after the big bang, so you could not have witnessed